

Problem-Solution Fit

Date	12/07/2026
Name	Karuru Arjun
Project Name	Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

1. CUSTOMER SEGMENT(S) [CS] - Students - Teachers & Faculty - Educational Institutions - Interview Candidates - Corporate Trainers - Online Learning Platforms - Researchers in AI & Education	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] - Poor communication skills - Language pronunciation variations - Lack of confidence while speaking - Limited access to expert evaluators - Difficulty receiving personalized feedback - Time constraints during assessments	5. AVAILABLE SOLUTIONS PROS & CONS [AS] - Manual Evaluation Pros: Human judgment, Detailed feedback Cons: Time-consuming, Subjective, Not scalable - Online Quiz Platforms Pros: Easy to use, Fast scoring Cons: Cannot evaluate spoken explanations, tests memory more than understanding - Speech Recognition Tools Pros: Accurate transcription Cons: No semantic understanding analysis, No speech fluency evaluation			
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] - Manual evaluation of spoken explanations is time-consuming (Very Frequent) - Subjective assessment by different evaluators (Frequent) - Difficulty identifying conceptual understanding accurately (Frequent) - Lack of personalized feedback on communication skills (Very Frequent) - No automated speech fluency analysis (Frequent) - Difficult to track student progress over time (Frequent)	9. PROBLEM ROOT / CAUSE [RC] - Manual assessment methods - Lack of intelligent evaluation systems - No semantic understanding measurement - Limited speech communication analysis - Absence of automated feedback generation - Difficulty measuring conceptual knowledge objectively	7. BEHAVIOR + ITS INTENSITY [BE] - Practice concept explanations regularly (High) - Upload audio for evaluation (High) - Review AI-generated feedback (High) - Download performance reports (Medium) - Improve speaking based on suggestions (High)			
3. TRIGGERS TO ACT [TR] - Students preparing for interviews - Online learning and assessments - Need for instant evaluation - Increasing use of AI in education - Teachers handling large numbers of students - Demand for personalized learning feedback	10. YOUR SOLUTION [SL] Develop an AI-powered Voice-Based Concept Understanding Analyser (VBCUA) using Python and Streamlit. The system converts spoken explanations into text using OpenAI Whisper, evaluates conceptual understanding through Sentence-BERT semantic similarity, analyzes speech fluency using Librosa, and generates intelligent scores with qualitative feedback. It also provides waveform visualization, downloadable PDF reports, and an interactive dashboard for students, educators, and researchers. Benefits - Automated concept understanding evaluation - AI-powered semantic similarity analysis - Speech fluency and confidence assessment - Instant qualitative feedback - Interactive Streamlit dashboard - Downloadable PDF performance reports - Scalable solution for educational institutions				
4. EMOTIONS BEFORE / AFTER [EM] <table border="0"> <tr> <td>BEFORE</td> <td>AFTER</td> </tr> <tr> <td> - Lack of confidence - Nervousness while speaking - Uncertainty about understanding - Fear of poor communication - No clear feedback </td> <td> - Improved confidence - Better conceptual clarity - Enhanced communication skills - Instant AI feedback - Motivation for continuous improvement </td> </tr> </table>	BEFORE	AFTER	- Lack of confidence - Nervousness while speaking - Uncertainty about understanding - Fear of poor communication - No clear feedback	- Improved confidence - Better conceptual clarity - Enhanced communication skills - Instant AI feedback - Motivation for continuous improvement	8. CHANNELS OF BEHAVIOR [CH] ONLINE - Streamlit Web Application - College LMS - Educational Websites - Online Learning Platforms - Cloud Deployment OFFLINE - Classroom Demonstrations - Laboratory Sessions - Academic Workshops - Training Programs
BEFORE	AFTER				
- Lack of confidence - Nervousness while speaking - Uncertainty about understanding - Fear of poor communication - No clear feedback	- Improved confidence - Better conceptual clarity - Enhanced communication skills - Instant AI feedback - Motivation for continuous improvement				